



9407 - Unveiling the Dusty Engine of Massive Galaxy Evolution with ALMA and JWST

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(1) BX610

ABSTRACT

Joint ALMA-JWST proposal - ID: 2025.1.01186.S

We propose an ALMA+JWST joint proposal to achieve sub-kpc scale rest-frame near-IR and far-IR imaging of two massive star-forming galaxies (SFGs) that show evidence for rapid, large-scale gas inflows. The Band 4 cold dust imaging available for one target (BX610) at 2 kpc resolution reveals remarkable substructures within its few kpc dusty center. JWST imaging that exists for only one target (K20-ID7) further exhibits a hidden massive red core, which is unveiled only at the NIRCam 4 micron, and have stellar populations that seem linked to the rapid gas inflow. All the elusive details in the dusty engines of these massive SFGs are critical new knowledge about massive galaxy evolution. These require a high-resolution, high-quality joint ALMA+JWST+VLT(IFU) program. We propose joint ALMA Band 9 dust continuum imaging at 0.1 arcsec (4.7 hours in total) and JWST NIRCam 1-4 micron imaging (1.6 hours in total) in these representative massive main sequence SFGs. Together with deep near-IR IFU, this will greatly push the frontier of our knowledge about physics in the core and disk of massive SFGs with exquisite data sets.

OBSERVING DESCRIPTION

We request JWST NIRCam F115W+F277W and F150W+F444W imaging for BX610 which do not have JWST coverage in the whole GO1/GTO/ERS programs. Our target is bright ($K \sim 21$ mag; Förster Schreiber et al. 2018, Table 3), thus BRIGHT2 readout (8 groups) and INTRAMODULEBOX-4 dithering are ideal for a high-quality imaging and PSF sampling with minimum impact of cosmic rays (following the JWST technical report, e.g., JWST-STScI-001721, and other shallow-imaging programs, e.g., COSMOS-Web, FRESCO direct imaging).

Proposal 9407 - Targets - Unveiling the Dusty Engine of Massive Galaxy Evolution with ALMA and JWST

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	BX610	RA: 23 46 9.4300 (356.5392917d) Dec: +12 49 19.21 (12.82200d) Equinox: J2000		
Comments: Category=Galaxy Description=[High-redshift galaxies]					

Proposal 9407 - Observation 1 - Unveiling the Dusty Engine of Massive Galaxy Evolution with ALMA and JWST

Observation	Proposal 9407, Observation 1										Tue Oct 14 07:00:17 GMT 2025
	Diagnostic Status: Warning										
Diagnostics	Observing Template: NIRCam Imaging										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	BX610	RA: 23 46 9.4300 (356.5392917d) Dec: +12 49 19.21 (12.82200d) Equinox: J2000								
Template	Comments: Category=Galaxy Description=[High-redshift galaxies]										
	Module		Subarray				Target Placement				
Dithers	ALL		FULL				Module Gap				
	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
Spectral Elements	1	INTRAMODULEBOX		4		STANDARD				1	
	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
Special Requirements	1	F115W	F277W	BRIGHT2	8	1	4	4	687.153	202547	
	2	F150W	F444W	BRIGHT2	8	1	4	4	687.153	202547	
	Offset 50.0 arcsec, 20.0 arcsec										